

XIAMEN
UNIVERSITY

COM102 – Programming Language C

Chapter 6: Structures and Strings (ii)- Lab

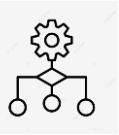
Dr. Najla'a A. M. Draib

In-Class Exercises

Note

- In C, **strings are usually char arrays + '\0'**.
- **Arrays (like** char name[50];) **cannot be assigned using = after declaration**.
- You can:
  - initialize them when you **declare them**:
    char name[50] = "John";
  - or **copy** into them:
    strcpy(name, "John");
  - use both **scanf** and **fgets** to store data into a string (a char array).
  - or fill them **character by character**.
- This rule is the same **inside a struct** as it is outside.

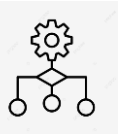
Exercise 1: Basic Structure Definition



- Define a **structure** named Book with the following members: title (string of 50 characters),author (string of 50 characters),price (float).

- Write a program to declare a variable of type Book, take input for all the fields, and print the details.

Exercise 1: Basic Structure Definition



- Define a **structure** named Book with the following members: title (string of 50 characters),author (string of 50 characters),price (float).

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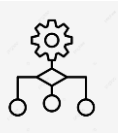
Exercise 2: Array of Structures

Define a structure named ***Employee*** with the following members:

- *id* (integer),
- *name* (string of 50 characters),
- *salary* (float).

Write a program to **declare an array of 5 Employee** structures. Input data for all employees and print their details

Exercise 3: Comparing StructuresDefine

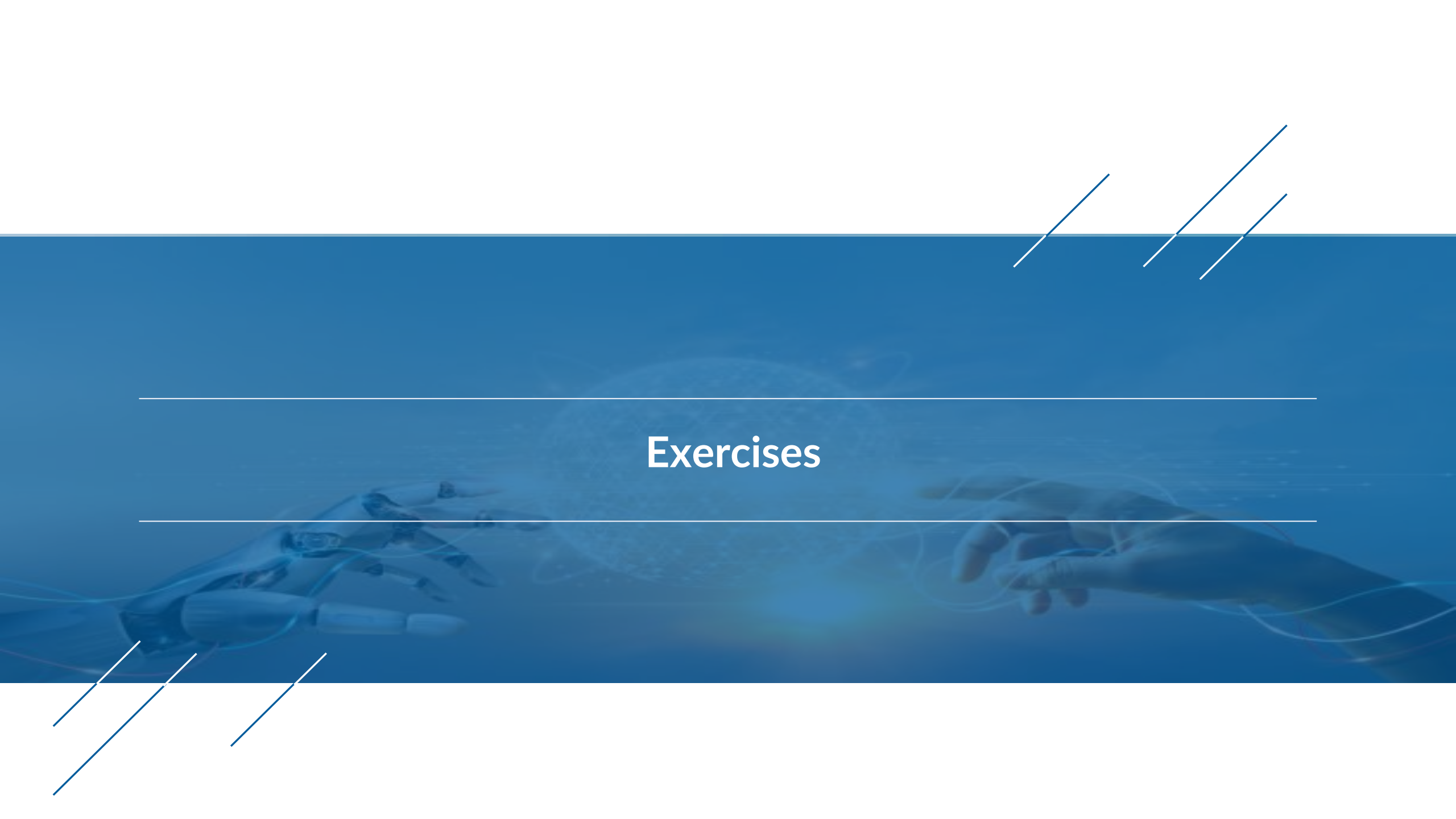


Define a structure named Person with:

- name (string),
- age (integer).

Write a program to:

- Create two variables of type Person.
- Compare their ages and print the name of the older person.



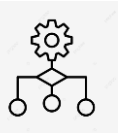
Exercises

Exercise 3: Pointer to Structure

Define a structure named ***Point*** with members **x** and **y** (both integers). Write a program to:

- Create a structure variable.
- Assign its address to a pointer.
- Use the pointer to modify and print the values of x and y.

Exercise 4: Passing Structures to Functions



Define a structure named Circle with:

- radius (float).

Write a program to:

  - Create a function calculateArea that takes a Circle structure as a parameter and returns the area.
  - Input a radius, pass the structure to the function, and display the area

Exercise 5: Array Inside a Structure

Define a structure named Student with:

- name (string),
- grades (an array of 5 integers),
- age (integer).

Write a program to input and display details of one student, including all 5 grades.

Exercise 6: Structure as Return Value

Define a structure named Rectangle with:

- length (float),
- breadth (float).

Write a function that:

- Takes length and breadth as inputs,
- Returns a Rectangle structure. In the main function, display the values of the returned structure

Exercise 7: Comparing StructuresDefine

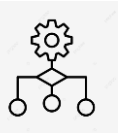
a structure named Person with:

- name (string),
- age (integer).

Write a program to:

- Create two variables of type Person.
- Compare their ages and print the name of the older person.

Exercise 9: Dynamic Memory Allocation for Structures



Define a structure named Movie with:

- name (string),
- rating (float).

Write a program to dynamically allocate memory for a Movie structure, input the details, and print them. Then free the allocated memory.

Exercise 10: Passing Structures to Functions

Define a structure named Circle with:

- radius (float).

Write a program to:

- Create a function calculateArea that takes a Circle structure as a parameter and returns the area.
- Input a radius, pass the structure to the function, and display the area.



Thank You

Q & A